Stock Price Prediction Web Application

Project Report

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Introduction

This project focuses on developing a machine learning-based web application for stock price prediction. The application analyzes historical stock market data and predicts short-term stock price trends using multiple machine learning algorithms.

The project aims to provide investors and users with meaningful insights into stock market behavior using data visualization and predictive analytics.

The system includes:

- Stock market trend analysis
- Technical indicator visualization
- Machine learning-based prediction
- Interactive web dashboard
- Real-time stock data visualization

Objectives

- To analyze historical stock market datasets.
- To build machine learning models for stock prediction.
- To visualize stock trends and technical indicators.
- To develop an interactive dashboard using Streamlit.
- To compare performance of multiple ML algorithms.

Tools and Technologies Used

Tool / Technology	**Purpose**
Python	Programming Language
Streamlit	Web Application Development
Pandas	Data Processing
Matplotlib	Data Visualization
Machine Learning Models	Stock Prediction
Financial APIs	Stock Data Collection
PyCharm	Development Environment

Machine Learning Models Used

The following machine learning algorithms were implemented and compared:

- Linear Regression
- Random Forest Regressor
- K-Nearest Neighbors (KNN)
- Extra Trees Regressor
- XGBoost Regressor

These models were trained using historical stock datasets to predict future stock price movements.

Features of the Application

- Real-time stock data analysis
- Interactive dashboard using Streamlit
- Technical indicator visualization
- Stock trend prediction
- Multiple ML model comparison
- User-friendly interface

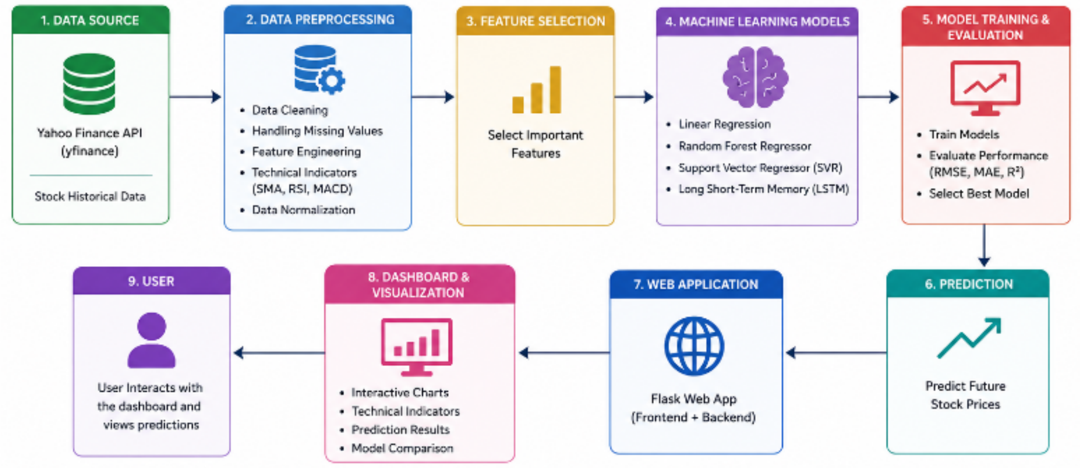
Technical Indicators Used

The application visualizes various technical indicators to help understand stock market behavior.

- RSI (Relative Strength Index)
- MACD (Moving Average Convergence Divergence)
- Bollinger Bands
- SMA (Simple Moving Average)
- EMA (Exponential Moving Average)

These indicators help identify market trends, momentum, and price volatility.

System Architecture



The system architecture includes data collection, preprocessing, model training, prediction generation, and dashboard visualization.

Project Workflow

1. Collect historical stock market data.
2. Perform data preprocessing and cleaning.
3. Calculate technical indicators.
4. Train machine learning models.
5. Compare prediction accuracy.
6. Display predictions and visualizations in Streamlit dashboard.

Advantages of the System

- Helps analyze stock market trends effectively.
- Provides interactive data visualization.
- Supports multiple machine learning models.
- Easy-to-use web interface.
- Improves understanding of predictive analytics.

Conclusion

The Stock Price Prediction Web Application successfully demonstrates the application of machine learning and data analytics in financial forecasting. The project improved understanding of data preprocessing, visualization, machine learning algorithms, and web application deployment.

The system provides meaningful stock market insights through predictive analytics and interactive dashboards.

Future Scope

- Add deep learning models such as LSTM.
- Integrate real-time live market prediction.
- Add portfolio analysis features.
- Improve prediction accuracy using advanced AI techniques.
- Deploy application on cloud platforms.

— **End of Report** —